

Bipolarons

A *bipolaron* is a composite quasi-particle comprising (1) two charge carriers bound within a common self-trapping potential well (2) taken together with the displaced atomic equilibrium positions that produce it. The self-trapped carriers of a *large* bipolaron encompass several elementary structural units. By contrast, a *small* bipolaron's two self-trapped carriers collapse to a single elementary structural unit. When the two carriers share a common state the Pauli principle requires that their spins assume an anti-parallel alignment to form a singlet. Furthermore, as illustrated in Fig. 7.1, the merger of more than two carriers is opposed by the requirement that additional carriers occupy higher lying states. All told, coalesced carriers usually are presumed to pair as singlet bipolarons.

The electron–phonon interaction fosters the merger of self-trapped carriers. In particular, polaron models show the depth of a self-trapping potential well doubling when it is occupied by two carriers. Thus, the self-trapping contribution to each carrier's potential energy is lowered by their merger. However, the tendency for self-trapped carriers to coalesce is opposed by their mutual Coulomb repulsion. If and when self-trapped carriers amalgamate depends upon the resolution of these competing effects.

This chapter addresses the conditions governing the formation of large and small bipolarons. This discourse emphasizes the roles of the long- and short-range electron–phonon interactions. With just the Fröhlich long-range electron–phonon interaction two equivalent uncorrelated carriers will be bound provided that the static dielectric constant is much larger than the high-frequency optical dielectric constant, $\epsilon_0 > 2\epsilon_\infty$. However, these carriers' bipolaron is not stable with respect to separation into two separate polarons. Nonetheless, stabilization of this bipolaron can be achieved by reducing the two carriers' mutual Coulomb repulsion or enhancing their electron–phonon interaction. The carriers' mutual Coulomb repulsion can be reduced by electron-correlation effects that keep the paired carriers away from



Fig. 7.1 The occupation of non-degenerate self-trapped states is depicted for the merger of two singlet bipolarons into a quad-polaron.

one another. The electron–phonon interaction can be enhanced with the addition of a short-range component. However, if the short-range component is too strong, a large bipolaron will collapse into a small bipolaron.

Hund’s rule states that the exchange energy between two carriers occupying degenerate states stabilizes their forming a triplet. However, if the carriers possess strong enough electron–phonon interactions with symmetry-breaking degeneracy-lifting atomic displacements a singlet will be stabilized instead. Firstly, coupling carriers’ energies to symmetry-breaking atomic displacements reduces their stiffness constants. The lowering of the concomitant vibrations’ free energy may be enough to stabilize the formation of a “softening” singlet bipolaron. Secondly, these electron–phonon interactions may be strong enough compared with the exchange energy to produce degeneracy-lifting shifts of atoms’ equilibrium positions which stabilize a singlet bipolaron. Such a bipolaron may be termed a Jahn–Teller bipolaron.

7.1 Large-bipolaron formation

The scaling approach of Chapter 4 is used to address the formation and stability of a singlet bipolaron having d -dimensional electronic carriers (Emin and Hillery, 1989). To begin, consider the energy functional for the bipolaron’s adiabatic ground state:

$$\begin{aligned}
 E_2(L) &= \frac{2T_e}{L^2} - \frac{4V_L}{L} - \frac{4V_S}{L^d} + \frac{U}{L} \\
 &= \frac{2T_e}{L^2} - \frac{U}{L} \left(1 - \frac{2\varepsilon_\infty}{\varepsilon_0} \right) - \frac{4V_S}{L^d}.
 \end{aligned}
 \tag{7.1}$$

This energy functional is obtained by extending the energy functional for a single self-trapped carrier, Eq. (4.30), to that for a pair of self-trapped carriers occupying a common state. In particular, having a pair of carriers (1) doubles their net kinetic energy, (2) quadruples their net polaronic energy since each carrier experiences double the polaronic energy lowering, and (3) adds the pair’s mutual Coulomb repulsion. Here Eq. (4.32) has been utilized to express V_L , the long-range Fröhlich contribution to the polaron energy, in terms of U , the mutual Coulomb

repulsion of two equivalent carriers immersed in a medium whose optical dielectric constant is ε_∞ :

$$V_L \equiv \frac{U}{2} \left(1 - \frac{\varepsilon_\infty}{\varepsilon_0} \right). \quad (7.2)$$

Establishing a minimum of $E_2(L)$ at an intermediate value of L corresponds to forming a large bipolaron. By contrast, small-bipolaron formation corresponds to the minimum of $E_2(L)$ collapsing to the smallest permitted value of L , L_c . Then the self-trapped carriers are confined to a single structural unit.

The electron–phonon interaction will be sufficient to bind two carriers into a bipolaron if the energy functional $E_2(L)$ becomes negative. Requiring that $E_2(L)$ of Eq. (7.1) be negative yields the condition for bipolaron formation:

$$\frac{2T_e}{L} < U \left(1 - \frac{2\varepsilon_\infty}{\varepsilon_0} \right) + \frac{4V_S}{L^{d-1}}. \quad (7.3)$$

In the absence of the short-range portion of the electron–phonon interaction ($V_S = 0$) binding can only occur if $\varepsilon_0 > 2\varepsilon_\infty$. This condition is not satisfied in covalent or common ionic materials but is fulfilled in materials whose especially displaceable ions make $\varepsilon_0/\varepsilon_\infty$ exceptionally large (e.g. in relaxor or conventional ferroelectrics).

As illustrated in Fig. 7.2, adding a progressively stronger short-range component of the electron–phonon interaction increases the binding of a $d = 3$ large bipolaron while decreasing its spatial extent. Moreover, when V_S becomes large enough the minimum of $E_2(L)$ collapses to L_c , indicating small-bipolaron formation.

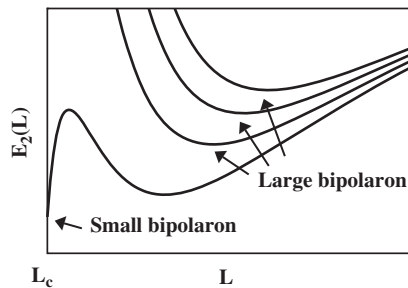


Fig. 7.2. The minimum of the bipolaron energy functional $E_2(L)$ at $L > L_c$, the smallest permitted value of L , corresponds to large-bipolaron formation. The large-bipolaron minimum deepens and its characteristic length decreases as the strength of the short-range component of the electron–phonon interaction V_S increases. The minimum of the energy functional ultimately shifts to L_c indicating the large bipolaron’s collapse into a small bipolaron.

Beyond being bound, a realizable bipolaron should be stable with respect to separation into two independent polarons. To appreciate the physical content of this requirement, the bipolaron energy functional of Eq. (7.1) is rewritten in terms of that for two separated polarons (the square bracketed term):

$$E_2(L) = 2 \left[\frac{T_e}{L^2} - \frac{V_L}{L} - \frac{V_S}{L^d} \right] + \frac{\epsilon_\infty}{\epsilon_0} \frac{U}{L} - \frac{2V_S}{L^d}. \quad (7.4)$$

Although the long-range electron–phonon interaction by itself produces a bound bipolaron if $\epsilon_0/\epsilon_\infty$ is sufficiently large, this $V_S = 0$ (Fröhlich) bipolaron remains unstable with respect to separation into two independent polarons (Pekar, 1963). In particular, the long-range electron–phonon interaction only reduces the net single-site repulsion between polarons from U to $(\epsilon_\infty/\epsilon_0)U$. This scaling-analysis result (Emin and Hillery, 1989) agrees with electrostatic considerations (Schultz, 1963).

It is evident from Eq. (7.4) that the addition of the short-range component of the electron–phonon interaction can stabilize a large bipolaron against its decomposition. However, the short-range electron–phonon interaction also fosters the collapse of a large bipolaron into a small bipolaron. The range of values of $2V_S/T_e$ for which a large bipolaron is stable against both decomposition and collapse widens as $\epsilon_0/2\epsilon_\infty$ increases beyond unity. For example, the range for which a planar ($d = 2$) large bipolaron is realizable is

$$\frac{4(\epsilon_0/2\epsilon_\infty) - 3}{2(\epsilon_0/2\epsilon_\infty)^2 - 1} < \frac{2V_S}{T_e} < 1. \quad (7.5)$$

The scaling approach has been used to study the energy of a pair of polarons as a function of their separation (Emin and Hillery, 1989). Even with stable bipolaron formation, the net energy of two polarons increases as their separation decreases until they overlap sufficiently to drive their merger into a bipolaron. In other words, there is an energy barrier to polarons coalescing into a bipolaron.

Bipolaron formation can involve correlating the positions of the two electronic carriers so that they can better avoid one another and thereby reduce their mutual Coulomb repulsion. However, keeping the two carriers away from one another also tends to reduce the polaronic binding that comes from the two carriers' merger.

Correlation effects are incorporated into a calculation of a bipolaron's energy by lifting the constraint that the bipolaron's two carriers occupy a common single-carrier orbital. For example, a correlated wavefunction, symmetric with respect to interchange of its two carriers, can be constructed with each carrier occupying a different single-carrier orbital. Correlated wavefunctions for one-dimensional large bipolarons were constructed by taking their two orbitals to have (1) different spatial

extents and (2) their centroids displaced from one another (Emin, Ye and Beckel, 1992). A correlated wavefunction for a planar large bipolaron was constructed from two elliptical orbitals with their major axes perpendicular to one another (Emin, 1995a). The resulting four-lobed large bipolaron is stabilized when its electronic bandwidth is large enough.

Variational calculations, such as the two cited above, are used to address these refinements of the competition between carriers' mutual Coulomb repulsion and their mutual attraction induced by their electron–phonon interactions. The results of such computations are sensitive to choices of variational wavefunctions (Vinetskii and Gitterman, 1958; Vinetskii, 1961; Suprun and Moizhes, 1982; Hiramoto and Toyozawa, 1985; Adamowskii, 1989; Verbist, Peters and Devreese, 1991; Basani, Geddo, Iadonisi and Ninno, 1991; Sil, Giri and Chatterjee, 1991). As such, qualitative features, rather than numerical results, appear most significant.

7.2 Small-bipolarons: negative- U centers

Charge carriers that collapse into small polarons readily localize, move by thermally assisted hopping, and typically manifest Curie paramagnetism. However, sometimes the expected Curie paramagnetism is not observed. One explanation of the absent Curie paramagnetism is that the carriers find it energetically favorable to localize as singlet pairs, small bipolarons, rather than as small polarons. Even then paired carriers tend to break apart as they hop to produce a relatively small thermally activated Curie paramagnetism (Emin, 1996).

Models of two carriers' collapse into a singlet small bipolaron often consider only the short-range electron–phonon interaction. Indeed, the long-range component of the electron–phonon interaction vanishes in non-polar materials where $\epsilon_0 \rightarrow \epsilon_\infty$. In this limit the energy functional for a singlet bipolaron, Eq. (7.1), becomes

$$E_2(L) = \frac{2T_e}{L^2} - \frac{4V_S}{L^d} + \frac{U}{L}. \quad (7.6)$$

The collapse of two carriers into a small bipolaron is described by this energy functional having its negative absolute minimum at the smallest acceptable value of L_c , the value of L corresponding to the two carriers merging at a single site. The condition for the collapse to a small bipolaron becomes: $(4V_S - U) > 4T_e/L_c$ for $d = 1$, $(4V_S - 2T_e)/L_c > U$ for $d = 2$, and $4V_S/(L_c)^3 > U/L_c + 2T_e/(L_c)^2$ for $d = 3$. All told, small-bipolaron formation requires (1) a strong-enough short-range electron–phonon interaction, (2) a sufficiently weak inter-carrier Coulomb repulsion, and (3) a sufficiently narrow electronic bandwidth. In the narrow-band limit, $T_e \rightarrow 0$, the requirement that a small bipolaron be bound can be rewritten simply as $4E_b > U_H$.

where $E_b \equiv V_S/(L_c)^d$ is Holstein's small-polaron binding energy and $U_H \equiv U/L_c$ is the single-site Coulomb repulsion (Holstein, 1959b).

The difference between the small-bipolaron energy functional and that for two well-separated small polarons can be written in terms of E_b and U_H :

$$E_2(L_c) - 2\left(\frac{T_e}{L_c^2} - E_b\right) = -2E_b + U_H. \quad (7.7)$$

It is evident from Eq. (7.7) that this small bipolaron is energetically stable with respect to separating into two small polarons if $2E_b > U_H$. This short-range attraction between the two small polarons has been described as their having an attractive Hubbard interaction and as their having a negative- U (Pincus, 1972; Anderson, 1975).

Evidence for small bipolarons primarily consists of (1) observing carriers' hopping transport together with (2) the failure to observe the Curie paramagnetism expected of unpaired localized carriers. Such measurements generally occur in ionic and polar materials that are characterized by strong long-range electron-phonon interactions. In these materials the effective Coulomb repulsion between carriers is diminished by small ratios of optical-to-static dielectric constants. In particular, small-bipolaron hopping is said to occur in an intermediate-temperature (~ 130 K to ~ 150 K) phase of single-crystal Ti_4O_7 (Lakkis *et al.*, 1976). These bipolarons are based on paired d -electrons of Ti^{3+} ions that are adjacent to one another. Hopping-type bipolarons are also reported in the low-temperature phase of slightly reduced crystalline WO_3 (Schirmer and Salje, 1980). Small bipolarons are also indicated in reduced- and doped- BaTiO_3 (Moizhes and Suprun, 1984; Kolodiazhnyi and Wimbush, 2006; Shao *et al.*, 2008).

Bipolaron formation is sometimes attributed to atoms of elements that manifest the "inert-pair effect". These atoms will not adopt valences that require breaking pairs of their outermost s -electrons. In particular, Tl can have valences of 1+ and 3+ but not 2+; Pb can have valences of 2+ and 4+ but not 3+; Bi can have valences of 3+ and 5+ but not 4+. For example, atoms of Bi in crystalline BaBiO_3 are idealized as alternating between Bi^{3+} and Bi^{5+} ions along the crystal's three principal directions. This structure is sometimes described as an ordered arrangement of Bi-based small bipolarons.

7.3 Softening bipolarons

At first glance boron carbides ($\text{B}_{12+x}\text{C}_{3-x}$ with $0.06 < x < 1.7$) appear as prime examples of materials whose carriers are small bipolarons. These materials' conductivities and Hall mobilities manifest thermally activated transport (Wood, 1986).

Moreover, the holes introduced by replacing carbon atoms with boron atoms do not produce Curie paramagnetism (Azevedo *et al.*, 1985). Furthermore, boron carbides also exhibit a small thermally activated paramagnetism attributable to breaking singlet pairs (Chauvet *et al.*, 1996). However, these materials do not display the customary polaron absorption, produced by exciting self-trapped carriers from the potential wells that bind them (Samara *et al.*, 1993). Moreover, the entropy transported with a charge carrier, measured by the Seebeck coefficient, is unusual as it is dominated by a very large contribution that is independent of the carrier concentration (Aselage *et al.*, 1998; 2001). The unusual absorption and Seebeck coefficient are suggestive of a “softening bipolaron”.

Boron carbides are mixed crystals comprising $B_{11}C$ icosahedra and three-atom chains (Emin, 2006a). Electrons are donated from the chains to the electron-deficient icosahedra. Altering a chain’s atomic constituents can change its donation to icosahedra thereby modifying the carrier density. Carriers move by thermally assisted hopping between icosahedra. The icosahedra are hollow spheroids whose frontier molecular orbitals have f -symmetry. An icosahedron’s crystal field splits the seven-fold orbital degeneracy of these f -symmetry molecular orbitals into a low-energy four-fold-degenerate manifold and a higher energy three-fold-degenerate manifold. Electron donation in boron carbides ($B_{12+x}C_{3-x}$ with $0.06 < x < 1.7$) nearly fills the four-fold-degenerate manifold. In particular, the maximum hole concentration occurs near $x = 1$ yielding a singlet-bipolaron hole on about half of the icosahedra and a filled four-fold manifold on the remaining icosahedra.

The four-fold orbital degeneracy and two-fold spin degeneracy for each hole permits formation of many, $(4 \times 2)[(4 \times 2) - 1]/2 = 28$, different bipolaron states. Of these two-carrier states: four are same-state singlets, six are mixed-state singlets and 18 comprise triplets. Degeneracy among these two-carrier states is partially lifted by carriers’ exchange interactions.

The concept of a softening bipolaron emerges upon investigating how a singlet bipolaron can be stabilized on an icosahedron (Emin, 2000b). In particular, as enunciated by Hund’s rule, two carriers occupying degenerate orbitals are driven by their exchange interaction to align as a triplet rather than as a singlet. Nonetheless, as will become evident, static and dynamic symmetry-breaking deformations can overcome the exchange interaction and stabilize a singlet.

To appreciate how a singlet can be stabilized consider the algebraically simple example of two-fold orbital degeneracy (Emin, 2000b). The six two-carrier states of this example, $(2 \times 2)[(2 \times 2) - 1]/2 = 6$, consist of three singlets plus the three states of a triplet. Imposing a symmetry-breaking atomic distortion characterized by the parameter z can lower the electronic energy of the lowest-lying singlet W_S below the electronic energy of the triplet manifold W_T :

$$(W_S - W_T) = \left[3J - \sqrt{J^2 + (4F_z z)^2} \right] / 2, \quad (7.8)$$

where J is the exchange splitting and F_z is the linear electron–phonon coupling constant associated with z . Without a deformation this singlet lies above the triplet: $W_S - W_T = J$ for $z = 0$. However, when the symmetry-breaking deformation is large enough the singlet state falls below the triplet, $W_S - W_T \rightarrow -2F_z z$ as $z \rightarrow \infty$.

That portion of the adiabatic potential which depends on symmetry-breaking is obtained upon adding its strain energy to its contribution to the lowest-lying singlet's electronic energy:

$$V_{\text{ad},z}(z) = \left[kz^2 + J - \sqrt{J^2 + (4F_z z)^2} \right] / 2 \\ = E_{\text{bp},z} \left\{ \left(\frac{kz}{2F_z} \right)^2 + \frac{2}{\gamma} \left[1 - \sqrt{1 + \gamma^2 \left(\frac{kz}{2F_z} \right)^2} \right] \right\}, \quad (7.9)$$

where $E_{\text{bp},z} \equiv 4E_{\text{p},z} \equiv 2F_z^2/k$ and $\gamma \equiv 4E_{\text{bp},z}/J$. Here $E_{\text{p},z}$ is the contribution to the localized polaron's binding energy from symmetry-breaking displacements of atoms' equilibrium positions. The symmetry-breaking contribution to the adiabatic potential $V_{\text{ad},z}(z)$ is plotted in Fig. 7.3 for several values of γ . The stiffness of the minimum at $z = 0$, $k(1 - \gamma)$, is progressively reduced as γ is increased from zero to

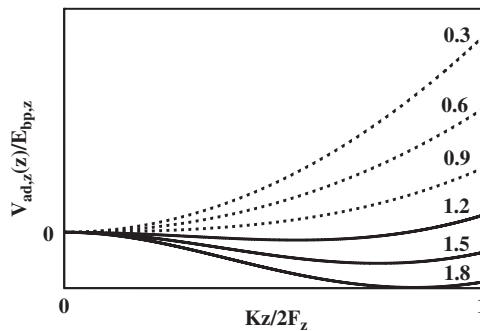


Fig. 7.3 The adiabatic potential for the lowest-lying singlet is plotted against the symmetry-breaking deformation parameter for six (indicated) values of the electron–phonon coupling parameter $\gamma \equiv 16E_{\text{p},z}/J$. Here $E_{\text{p},z}$ is the contribution to the localized polaron's binding energy from symmetry-breaking displacements of atoms' equilibrium positions and J is the singlet–triplet exchange splitting in the absence of distortion. The potential about the minimum at $z = 0$ softens as γ is increased toward unity. By contrast, the minimum shifts, deepens and stiffens as γ is increased above unity.

unity. As γ is increased beyond unity the minimum shifts to $z = (2F_z/k)(1 - 1/\gamma^2)^{1/2}$ as it concomitantly deepens to $-E_{\text{bp},z}(1 - 1/\gamma^2)$ and stiffens to $k(1 - 1/\gamma^2)$.

These carrier-induced symmetry-breaking alterations of atomic vibrations change the coupled system's free energy by

$$F(\gamma, \omega) \equiv \kappa T \ln \left\{ \frac{\sinh[(\hbar\omega/2\kappa T)\sqrt{1-\gamma}]}{\sinh(\hbar\omega/2\kappa T)} \right\} \quad (7.10)$$

for $\gamma < 1$ and by

$$F(\gamma, \omega) \equiv -\frac{J}{4} \frac{(\gamma-1)^2}{\gamma} + \kappa T \ln \left\{ \frac{\sinh[(\hbar\omega/2\kappa T)\sqrt{1-1/\gamma^2}]}{\sinh(\hbar\omega/2\kappa T)} \right\} \quad (7.11)$$

for $\gamma > 1$. Figure 7.4 plots $F(\gamma, \omega)$ in units of $\hbar\omega$ against γ . It is evident that the free energy has minima at $\gamma = 1$ and falls with increasing γ when $\gamma \gg 1$. The free-energy minimum at $\gamma = 1$ is driven by lowering of the vibration frequency associated with symmetry-breaking atomic displacements. As such this softening minimum deepens with rising temperature. By contrast, the reduction of the free energy at $\gamma \gg 1$ results primarily from temperature-independent symmetry-breaking displacements of atoms' equilibrium positions. Higher electronic degeneracy and additional symmetry-breaking vibration modes enhance softening effects, broadening the softening region beyond that of this minimalist example (Emin, 2000b).

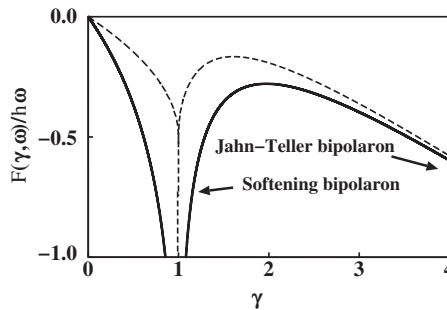


Fig. 7.4 $F(\gamma, \omega)/\hbar\omega$ is plotted against the electron-phonon coupling constant γ at two temperatures: $\kappa T = \hbar\omega/20$ (dashed curve) and $\kappa T = \hbar\omega$ (solid curve). The minimum of the free energy centered at $\gamma = 1$ primarily arises from lowering the frequency of the symmetry-breaking vibration. The bipolaron is dubbed a softening bipolaron when this lowering of vibrations' free energy is decisive in stabilizing it as a singlet. The fall of the free-energy at $\gamma \gg 1$ primarily arises from a symmetry-breaking shift of atoms' equilibrium positions. A Jahn-Teller bipolaron is stabilized primarily by symmetry-breaking shifts of atoms' equilibrium positions.

To stabilize a bipolaron formed among degenerate orbitals as a singlet rather than as a triplet requires that $-F(\gamma, \omega) > J$. The bipolaron is termed a softening bipolaron if its stabilization as a singlet results primarily from lowering atoms' vibration frequencies. By contrast, stabilization of a conventional or Jahn–Teller singlet bipolaron results mainly from shifting atoms' equilibrium positions.

The bipolaron will be energetically stable against separating into two polarons if

$$2E_p + [-F(\gamma, \omega) - 2E_{p,z}] > U, \quad (7.12)$$

where E_p and $E_{p,z}$ are the respective contributions to a polaron's binding energy from symmetry-conserving and symmetry-breaking displacements of atoms' equilibrium positions. In the absence of symmetry-breaking vibrations Eq. (7.12) simply reproduces the condition described below Eq. (7.7). The form of this condition is also reproduced in the strong-coupling limit $\gamma \gg 1$, where $-F(\gamma, \omega) \rightarrow E_{bp,z} = 4E_{p,z}$. In the complementary regime of Fig. 7.4 the reduced vibration frequencies of a singlet softening bipolaron, $\gamma = 16E_{p,z}/J \sim 1$ with $-F(\gamma, \omega) > J$, bolster the bipolaron's stability against decomposition into separate polarons since $-F(\gamma, \omega) \gg 2E_{p,z}$.

Distinctive features of boron carbides favor its charge carriers forming softening bipolarons (Emin, 2006a). In essence U is small (≈ 0.15 eV) because of electronic screening, $\epsilon_\infty \approx 10$, and the bipolarons' two holes being confined to the surface of an icosahedron whose radius is large (~ 1.8 Å) and about 1/3 of carriers' mean separation. Conventional polaronic binding is also greatly reduced by carriers being spread over an icosahedron's 20 faces. Furthermore, softening effects are enhanced by the large size and four-fold orbital degeneracy of an icosahedron's frontier orbitals.

As will be discussed in Section 9.1 the optically assisted transfer of one of a small-bipolaron's electronic carriers to an unoccupied site produces an absorption band centered at $h\nu = 4E_p - U$ for $4E_p > U$ (Emin, 1993). This condition is always satisfied for a small bipolaron since its stability requires that $2E_p > U$. However, such absorption need not occur for a softening bipolaron since its stability is primarily driven by lowering atoms' vibration frequencies, the square-bracketed term of Eq. (7.12). In particular, a stable softening bipolaron will not produce an absorption band if:

$$4E_p < U < 2E_p + [-F(\gamma, \omega) - 2E_{p,z}], \quad (7.13)$$

where $-F(\gamma, \omega) \gg 2E_{p,z}$ for $\gamma \approx 1$. Thus, the absence of a polaron absorption peak in boron carbides is consistent with its carriers being softening bipolarons.

As detailed in Section 12.3 an electronic carrier's softening of atoms' vibrations produces a contribution to its Seebeck coefficient which is independent of the carrier concentration (Emin, 1999). Measurements of boron carbides' Seebeck coefficients find a very large contribution of this type (Aselage *et al.*, 1998; 2001). This distinctive finding is also consistent with boron carbides' carriers forming softening bipolarons.